

XINGYU LIU

xliufs@connect.ust.hk (+852) 5109 9729

EDUCATION

The Hong Kong University of Science and Technology (HKUST), Hong Kong, China

Fourth-year Ph.D. Student in Electronic and Computer Engineering (ECE)

Sept. 2023 – Present

- Advisor: **Prof. Yansong Yang**
- CGA: **3.925/4.3**
- Qualifying Exam: *Passed on Nov. 13, 2024*

Harbin Institute of Technology, Shenzhen (HITSZ), Shenzhen, China

B.E. in Communication Engineering

Sept. 2019 – Jun. 2023

- Overall GPA: **92.498/100**
- In-school Ranking: **6/188**
- Outstanding Undergraduate Thesis (HITSZ)

RESEARCH EXPERIENCE

6G Radio Frequency Filters for Mobile Devices (*Innovation and Technology Fund*)

Participating Project Team (HKUST); Lead: Prof. Yansong Yang

2025 – 2026

- Conducted MEMS/mmWave acoustic filter design and validation for next-generation mobile RF front-ends.
- Related work was published as first-author or co-author papers in *IEEE TMTT* and *IEEE UFFC-JS*.

Technology for Next-Generation Wide-/Ultrawide-Bandgap Semiconductor Integrated Electronics (*RGC Strategic Topics Grant*)

Participating Project Team (HKUST); Lead: Prof. Kevin Jing Chen

2024 – 2028

- Performed wide-/ultrawide-bandgap device/process co-design and RF integration research.
- Related work was published in the *IEEE MEMS* conference proceedings.

New Methods for Suppressing Spurious Acoustic Modes in RF Micro-Acoustic Resonators/Filters (*NSFC*)

Participating Project Team (HKUST); Lead: Prof. Yansong Yang

2024 – 2026

- Developed methods for spurious-mode identification and suppression in high-frequency acoustic resonators.
- Related work was published in *IEEE TUFFC*, *IEEE UFFC-JS*, and *IEEE EDL*.

Monolithic Hetero-Integration of a GaN/AlScN Microsystem Chip for RF Applications (*Innovation and Technology Fund*)

Participating Project Team (HKUST); Lead: Prof. Yansong Yang

2023 – 2025

- Investigated GaN/AlScN hetero-integration and RF microsystem co-design.
- Related work was published in the *IEEE MEMS* conference proceedings.

Microwave/mmWave Micro-Acoustic Resonator and Filter Technology for Next-Generation Wireless Communications (*RGC Early Career Scheme*)

Participating Project Team (HKUST); Lead: Prof. Yansong Yang

2022 – 2025

- Developed miniaturized acoustic resonators and filters for 5G/6G systems, including device measurement and modeling.
- Related work was published in *IEEE UFFC-JS*.

Microfabrication and Modeling of On-Chip Inductors | HITSZ & HKUST

Research Advisors: Prof. K. X. Wang (HITSZ), Prof. Y. Yang (HKUST)

Feb. 2022 – Jun. 2023

- Conducted high-Q suspended-inductor modeling and EM-layout co-design; completed lithography-mask design and process planning.
- This work received the *Outstanding Undergraduate Thesis* award at HITSZ.

Miniaturized Balun-Filter via Capacitive Coupling | HITSZ

Research Advisor: Prof. K. X. Wang (HITSZ)

Feb. 2022 – May 2022

- Proposed a balun-bandpass integrated structure with improved stopband performance and reduced footprint; published a first-author paper at IEEE ISAP.

Wi-Fi Coverage Modeling & L2/L3 Packet Parsing | Huawei (Songshan Lake)

Software Intern, Data Communication Protocol Department

Jun. 2022 – Aug. 2022

- Visualized user-side Wi-Fi coverage and optimized indoor coverage, improving deployment efficiency and effective service area.

Wideband Circularly Polarized mmWave Antenna & Polarizer | HITSZ

Research Advisor: Prof. K. X. Wang (HITSZ)

Sept. 2021 – Dec. 2021

- Developed a magneto-electric dipole source and multilayer dielectric polarizer with enhanced gain and axial-ratio bandwidth; the work led to two authorized utility patents.

THD Measurement System for Small-Signal Testing | HITSZ

Project Member

Nov. 2020 – Nov. 2021

- Designed a THD measurement chain with mobile visualization; won Third Prize in the National Undergraduate Electronic Design Competition.

INDEPENDENT RESEARCH DIRECTION

- Initiated an independent research track in **AI-assisted MEMS device design**, developing intelligent models that identify and mitigate parasitic effects and recognize spurious responses in resonators and filters.
- Integrated vector fitting, deep neural networks, prompt-to-structure models, and multi-objective optimization to connect device geometry with simulated and measured responses.
- The long-term goal is to bring **intelligent manufacturing** into every stage of micro- and nanofabrication—from device design and process planning to fabrication monitoring, wafer-level testing, and production feedback.
- This direction aims to enable intelligent, modern, and scalable mass production of reliable MEMS devices for communications, 6G, aerospace, and other emerging applications.

PUBLICATIONS

First-Author Journal Papers

1. **X. Liu**, T. Meng, S. Chen, T. H. Yang, J. Zhang, S. Gong, and Y. Yang, “Toward intelligent design and measurement of MEMS acoustic wave resonators,” *IEEE Trans. Microwave Theory Tech.*, 2026, doi: 10.1109/TMTT.2025.3649908.
2. **X. Liu**, J. Zheng, and Y. Yang, “Scale interdigital transducer-based microacoustic resonators into mmWave applications,” *IEEE Trans. Ultrason., Ferroelectr., Freq. Control*, vol. 72, no. 5, pp. 674–685, May 2025, doi: 10.1109/TUFFC.2025.3554004.

First-Author Conference Papers

3. **X. Liu**, J. Zheng, F. Qian, Z. Ren, K. Yang, and Y. Yang, “IDT-based mmWave resonator with large impedance ratio and over-100% frequency offset between resonance and anti-resonance,” in *Proc. IEEE MTT-S Int. Microwave Symp. (IMS)*, 2026. Presented.

4. **X. Liu**, J. Zheng, F. Qian, Z. Ren, K. Yang, S. Gong, J. Zhang, and Y. Yang, "Intelligent resonator design framework integrating vector fitting, spurious score learning with DNN, and Pareto optimization for spurious-free high-Q performances," in *Proc. IEEE Int. Conf. Micro Electro Mechanical Systems (MEMS)*, Salzburg, Austria, 2026, doi: 10.1109/MEMS64181.2026.11419395.
5. **X. Liu**, J. Zheng, S. Chen, W. Wei, F. Qian, Z. Ren, K. Yang, and Y. Yang, "Intelligent prediction of EM responses in mmWave acoustic resonators via prompt-to-structure large language model and regression models," in *Proc. IEEE Int. Ultrasonics Symp. (IUS)*, Utrecht, Netherlands, 2025, pp. 1–4, doi: 10.1109/IUS62464.2025.11201527.
6. **X. Liu**, J. Zheng, Z. Ren, F. Qian, J. Xu, and Y. Yang, "Toward interdigital transducer-based mmWave acoustic: Mitigating self-resonances," in *Proc. IEEE Ultrason., Ferroelectr., Freq. Control Joint Symp. (UFFC-JS)*, Taipei, Taiwan, 2024, pp. 1–4, doi: 10.1109/UFFC-JS60046.2024.10794077.
7. **X. Liu**, Z. Wu, and K. Wang, "A miniaturized balun with bandpass responses and wide stopband," in *Proc. IEEE Int. Symp. Antennas Propag. (ISAP)*, Kuala Lumpur, Malaysia, 2023, pp. 1–2, doi: 10.1109/ISAP57493.2023.10388516.

Co-Author Journal Papers

8. F. Qian, S. Chen, W. Wei, J. Xu, K. Yang, J. Zheng, Z. Ren, **X. Liu**, and Y. Yang, "Suppressing acoustomigration and temperature rise for high-power robust acoustics," *Nature Communications*, 2026.
9. J. Xu, Z. Ren, F. Qian, S. Chen, W. Wei, J. Zheng, **X. Liu**, and Y. Yang, "A wafer-level sealed silicon cavity microacoustic platform for radio frequency integration," *Microsystems & Nanoengineering*, 2025.
10. J. Zheng, Z. Ren, J. Xu, **X. Liu**, F. Qian, and Y. Yang, "Periodically poled piezoelectric single-layered and multilayered lithium niobate for thickened high-order Lamb wave acoustic devices," *IEEE Trans. Microwave Theory Tech.*, 2025, doi: 10.1109/TMTT.2025.3567329.
11. J. Zheng, **X. Liu**, and Y. Yang, "Metal-air-piezoelectric (MAP) mmWave resonator," *IEEE Electron Device Lett.*, vol. 46, no. 2, pp. 167–170, Feb. 2025, doi: 10.1109/LED.2024.3513410.

Co-Author Conference Papers

12. S. Chen, F. Qian, Z. Ren, K. Yang, W. Wei, J. Zheng, J. Xu, **X. Liu**, X. Duan, and Y. Yang, "Layered acoustic wave devices with enhanced power handling based on lithium tantalate on silicon," in *Proc. IEEE Int. Conf. Micro Electro Mechanical Systems (MEMS)*, 2026.
13. J. Zheng, **X. Liu**, Z. Ren, K. Yang, F. Qian, J. Xu, and Y. Yang, "Enhancing mechanical stability of MEMS resonator at cryogenic temperature using periodically poled thin films," in *Proc. IEEE Int. Ultrasonics Symp. (IUS)*, Utrecht, Netherlands, 2025, pp. 1–4.
14. Z. Ren, J. Xu, F. Qian, J. Zheng, X. Duan, **X. Liu**, K. Yang, C. Zuo, H. Sun, and Y. Yang, "Pop-down FBAR structure for thermal stability without sacrificing coupling," in *Proc. IEEE Int. Ultrasonics Symp. (IUS)*, Utrecht, Netherlands, 2025, pp. 1–4.
15. F. Qian, S. Chen, W. Wei, K. Yang, J. Xu, J. Zheng, **X. Liu**, Z. Ren, and Y. Yang, "Boost the acoustic wave resonator to handle ultra-high power density beyond 10 W/mm²," in *Proc. IEEE Int. Ultrasonics Symp. (IUS)*, Utrecht, Netherlands, 2025, pp. 1–4. *Best Student Paper Poster Finalist*.
16. F. Qian, Z. Ren, **X. Liu**, J. Zheng, J. Xu, and Y. Yang, "Double-sided heat dissipation for acoustic resonators based on lithium niobate on sapphire," in *Proc. IEEE Int. Conf. Micro Electro Mechanical Systems (MEMS)*, Kaohsiung, Taiwan, 2025, pp. 1165–1168, doi: 10.1109/MEMS61431.2025.10918221.
17. J. Zheng, F. Qian, Z. Ren, J. Xu, **X. Liu**, and Y. Yang, "Temperature-driven degradations on lithium niobate MEMS acoustic filter," in *Proc. IEEE Ultrason., Ferroelectr., Freq. Control Joint Symp. (UFFC-JS)*, Taipei, Taiwan, 2024, pp. 1–4, doi: 10.1109/UFFC-JS60046.2024.10793653.
18. Z. Ren, F. Qian, **X. Liu**, J. Zheng, J. Xu, and Y. Yang, "Experimental study of the acoustic-electromagnetic hybrid filter at a wide temperature range," in *Proc. IEEE Ultrason., Ferroelectr., Freq. Control Joint Symp. (UFFC-JS)*, Taipei, Taiwan, 2024, pp. 1–3, doi: 10.1109/UFFC-JS60046.2024.10794075.

PATENTS

1. K. Wang, L. Qin, and **X. Liu**, "A Broadband Circularly Polarized Millimeter-Wave Antenna with Gain Enhancement." Authorized Publication No. CN218385766U; Application No. 202222912649.3; authorized Jan. 24, 2023.
2. K. Wang, X. Cai, and **X. Liu**, "A Low-Profile Circularly Polarized Fresnel Lens Antenna in the mmWave Band." Authorized Publication No. CN219040733U; Patent No. ZL202223574540.X; authorized May 16,

2023.

ACADEMIC AWARDS & SCHOLARSHIPS

- Best Student Paper Poster Finalist, IEEE International Ultrasonics Symposium (IUS), 2025
- HKUST RedBird PhD Award (AY 2024–25)
- HKUST RedBird PhD Award (AY 2023–24)
- Outstanding Undergraduate Thesis, Harbin Institute of Technology, Shenzhen, 2023
- National Scholarship (AY 2020–21)
- Third Prize, National Undergraduate Electronic Design Competition, Dec. 2021
- Third Prize, National College Students ROBOCON, Jul. 2021
- University First-Class Scholarship (AY 2019–20, 2020–21, and 2021–22)
- First Prize, National College Students Mathematics Competition, Dec. 2020
- Second Prize, National College Students Mathematical Contest in Modeling, Nov. 2020
- Enterprise Progress Scholarship (AY 2019–20)

LEADERSHIP & MENTORING EXPERIENCE

- Mentored junior students in MEMS/mmWave acoustic-resonator experiments, including RF device measurement, data analysis, equivalent-circuit modeling, and interpretation of resonator/filter responses.
- Guided a junior student toward a research publication, supporting experimental design, measurement validation, data organization, and manuscript preparation.
- Currently mentor an MPhil student in RF/MEMS device characterization, experiment planning, and research-problem formulation.
- Coordinate experimental workflows within the group, including measurement setup, troubleshooting, and knowledge sharing with junior group members.